

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming With Java

SESSION: 2025-26



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Abstract

The Customer Feedback Analysis System is a Java-based application developed to analyze and interpret customer reviews efficiently using Object-Oriented Programming (OOP) concepts. In today's digital environment, businesses receive a large volume of feedback in the form of ratings and comments, making manual analysis time-consuming and inefficient. This system automates the process by collecting multiple customer feedback entries, each consisting of a numerical rating and a textual comment.

The application uses a Feedback class to represent individual reviews and an Analyzer class to process the data. The system calculates the average rating from all feedback entries and categorizes the overall customer opinion into four levels: Excellent, Good, Average, and Poor. This classification is performed using switch-case logic, ensuring structured decision-making. The project demonstrates key programming concepts such as classes, objects, encapsulation, arrays, and control statements. It provides a simple yet effective solution for understanding customer satisfaction trends. The system can be further enhanced with advanced features like real-time data input, database integration, and graphical user interfaces, making it more scalable and practical for real-world applications.

Introduction

In today's digital world, customer feedback is an important factor for improving products and services. Businesses receive large amounts of feedback in the form of ratings and comments, which makes manual analysis difficult, time-consuming, and sometimes inaccurate. To overcome this problem, the Customer Feedback Analysis System is developed using Java and Object-Oriented Programming (OOP) concepts.

This system helps in collecting and analyzing multiple customer feedback entries efficiently. Each feedback consists of a rating score and a comment, which are stored using objects. The system then processes all feedbacks to calculate the average rating and determines the overall customer opinion. Based on the average score, the feedback is categorized into Excellent, Good, Average, or Poor using switch-case logic.

The project demonstrates the use of classes, objects, arrays, and decision-making techniques. It reduces manual effort and provides a simple and effective way to understand customer satisfaction levels.

Scope

This project is suitable for small libraries and can be extended with database integration in future.

System Requirements

Hardware: Computer with 4GB RAM

Software: Java JDK, IDE (VS Code/Eclipse/Edit Plus)

Methodology

1. Architecture

- **Model (Feedback):** Stores rating (int) and comment (String).
 - **Service (Analyzer):** Computes averages and maps sentiment.
 - **Controller (main):** Handles CLI input/output.
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2. Execution Flow

- **Allocation:** Initializes dynamic array size from input.
 - **Buffer Management:** Uses `sc.nextLine()` to prevent input skipping.
 - **Aggregation:** Computes precision average using double casting.
 - **Classification:** Converts average to label via `Math.round()`.
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3. Sentiment Boundaries

- **Excellent:** ≥ 4.5
- **Good:** ≥ 3.5 and < 4.5
- **Average:** ≥ 2.5 and < 3.5
- **Poor:** ≥ 1.5 and < 2.5
- **Bad:** < 1.5

If you want to upgrade this codebase, I can provide the quick implementation. Would you like to:

- Add **input validation** to prevent application crashes
- Convert the array to a dynamic **ArrayList**
- Implement a **text-based keyword filter** for comments

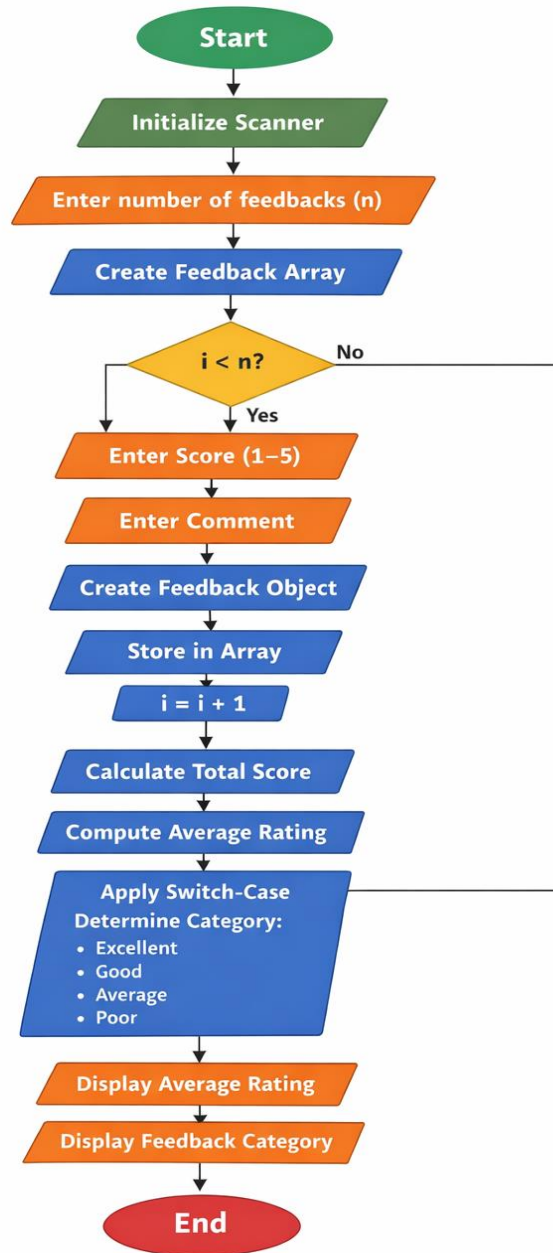
CLASS NAME

Variables inside class

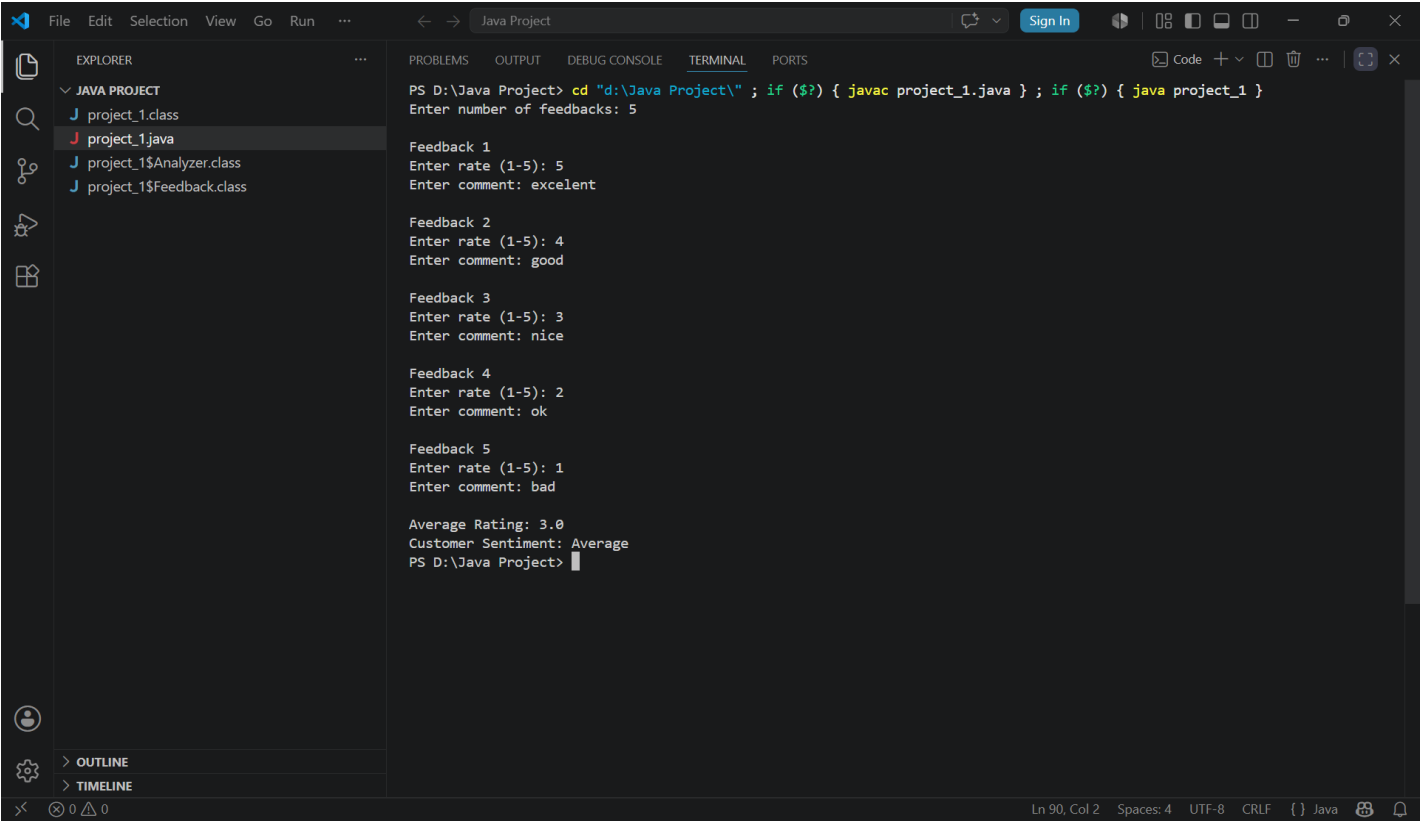
o Methods inside class

o If array is present within the class also mention it here

Customer Feedback Analysis System



SAMPLE INPUT / OUTPUT



Implementation

Explain different classes used in your program(only explain logic don't paste code here)

Sample Input/Output

Paste screenshot of your terminal with different operations

Future Enhancements

Things you can improve on your project

Gantt Chart

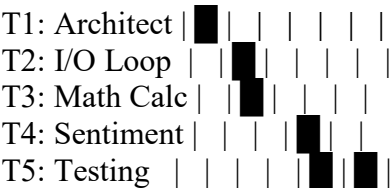
1. Project Timeline

Task	Duration	Phase
T1: Architecture	1 Day	Define Class Structures
T2: Console I/O	1 Day	Setup Scanner & Buffers
T3: Aggregation	1 Day	Code Average Calculations
T4: Sentiment	1 Day	Build Switch Mapping
T5: Testing	2 Days	Run Boundary QA

2. Timeline Visualization

text

DAYS | 1 | 2 | 3 | 4 | 5 | 6 |



Use code with caution.

If you want to view this visually using diagram tools, I can help. Would you like to:

- Get the Mermaid.js diagram code
- Add resource assignments to each task

Conclusion

The project successfully demonstrates Java OOP concepts and provides a functional system.

References

Book

Format:

[1] Author(s), *Book Title*, Edition. City: Publisher, Year.

Example:

[1] H. Schildt, *Java: The Complete Reference*, 11th ed. New York: McGraw-Hill, 2018.

Website

Format:

[3] Author (if any), "Title of page," Website Name. [Online]. Available: URL

Example:

[3] Oracle, "Java Documentation," Oracle. [Online]. Available: <https://docs.oracle.com>

Conference Paper

Format:

[4] Author(s), "Paper title," in Proc. Conference Name, year, pp.

Example:

[4] S. Kumar, "AI in Education," in Proc. IEEE Conf. on AI, 2021, pp. 120-125.

Journal Article

Format:

[2] Author(s), "Title of paper," Journal Name, vol., no., pp., year.

Example:

[2] A. Sharma and R. Gupta, "Machine Learning Techniques," International Journal of AI, vol. 5, no. 2, pp. 45-50, 2020.